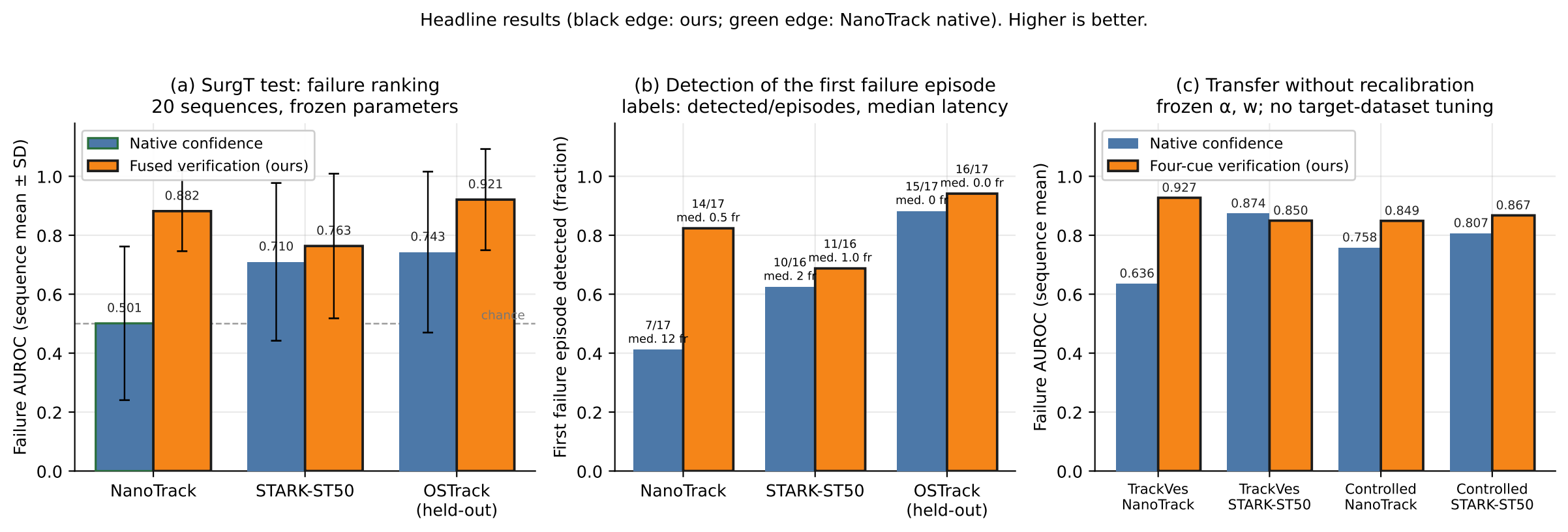
Headline results (black edge: ours; green edge: NanoTrack native). Higher is better.
(a) SurgT test: failure ranking
20 sequences, frozen parameters
Native confidence
Fused verification (ours)
Failure AUROC (sequence mean ± SD)
0.501
0.882
0.710
0.763
0.743
0.921
chance
NanoTrack
STARK-ST50
OSTrack
(held-out)
(b) Detection of the first failure episode
labels: detected/episodes, median latency
First failure episode detected (fraction)
7/17
med. 12 fr
14/17
med. 0.5 fr
10/16
med. 2 fr
11/16
med. 1.0 fr
15/17
med. 0 fr
16/17
med. 0.0 fr
NanoTrack
STARK-ST50
OSTrack
(held-out)
(c) Transfer without recalibration
frozen α, w; no target-dataset tuning
Native confidence
Four-cue verification (ours)
Failure AUROC (sequence mean)
0.636
0.927
0.874
0.850
0.758
0.849
0.807
0.867
TrackVes
NanoTrack
TrackVes
STARK-ST50
Controlled
NanoTrack
Controlled
STARK-ST50